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Invitation to Review for IEEE Access Access-2024-46290

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Sat, Jan 4, 2025 at 1:37 AM

Reply-To: ieeeaccess2@ieee.org

To: khare.pranav@gmail.com

04-Jan-2025

Dear Mr. Khare:

The manuscript entitled "BP-Vot: Blockchain-based e-Voting using Smart Contracts, Differential Privacy, and Self-Sovereign Identities" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Mehrdad Saif
Editor-in-Chief, IEEE Access

ieeeaccess2@ieee.org

MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: BP-Vot: Blockchain-based e-Voting using Smart Contracts, Differential Privacy, and Self-Sovereign Identities

AUTHORS: Baniata, Hamza; CALUNA, GIOVANNY

ABSTRACT: *Election is the key process typically utilized for maintaining democracy in a given society. Recent technological advancements, such as Blockchain (BC), have been already deployed in previous works to realize non-conventional e-Voting systems. The main goal for such proposals is to provide the necessary level of security and reliability, while maintaining transparency, trust, and remote elections. However, the distributed and publicity nature of BC brought new challenges related to privacy and performance trade-off. This paper aims to address these issues by integrating smart contracts for reliability and transparency, Differential Privacy for enhancing vote anonymity, and Self-Sovereign Identities to unlock the potential of the Web3 framework for verifiable credentials and digital identities. Specifically, a novel (k, ϵ) -differential privacy mechanism is developed, where a randomly selected candidate is pivoted from which retrievable votes are transferred to other candidates. Final election results are then statistically approximated. We evaluate the proposed methods for different arrival rates (10–80 TX/s), different total numbers of cast votes (10k–50k votes), and different numbers of elected candidates (2–8 candidates). To demonstrate the applicability of our proposal in real-life scenarios, we deploy our SC on a cloud-based permissioned BC network using Hyperledger Besu, with nodes set in Google's EU and USA data centers. Our experimental results showed that BP-Vot could provide 24% enhancement in latency over state-of-the-art solutions (≈ 1 s/TX compared to 1.24 s/TX). Additionally, using a standardized Min-Max regression mechanism, we show that BP-Vot could provide no less than 98% accuracy in votes approximation during all experiments, with a linearly increasing accuracy trend as a function of the total number of cast votes.*

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